

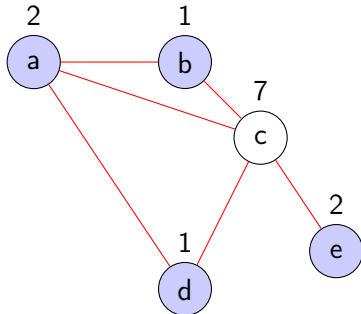
COMP1324 Algorithmic Linear Programming Rounding

Hikmat Farhat

- Linear programming rounding is a general method for finding approximate solution to hard problems.
- The idea is to relax the integrality constraints on the variables
- This allows us to efficiently solve the relaxed problem using LP.
- The solution to the relaxed problem could have non-integral values for the variables.
- The challenge is to **round** those values to obtain an approximate solution to the original problem.

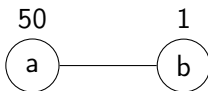
Example: weighted vertex cover

- In the weighted vertex cover problem each vertex $v \in V$ has a weight $w(v)$.
- The goal is to find a vertex cover C such that the total weight of the vertices in C is minimised instead of minimising the number of vertices.
- For example, in the instance below $\{a, b, d, e\}$ is now the optimal solution instead of $\{a, c\}$.



Modified approximation algorithm

- One can see why the previous algorithm is not suitable for the weighted vertex cover problem.
- It is no longer an α -approximation algorithm, for any α .
- Consider the instance below, the modified algorithm would select $\{a, b\}$ for a cost of 51 whereas the optimal value is 1.
- One can see that the solution can be made arbitrarily bad by increasing the weight of a .



- We can formulate the weighted vertex cover problem as a **integer** linear programming problem
- This is done by introducing a decision variable x_v for each vertex $v \in V$.
- We interpret $x_v = 1$ to mean that vertex v is in the vertex cover and $x_v = 0$ otherwise.
- The objective is to minimise the total weight of the vertices in the vertex cover.
- Therefore the objective function to be minimised is:

$$z = \sum_{v \in V} w(v) \cdot x_v$$

Constraints

- The constraints are as follows:

$$(x_u = 1) \vee (x_v = 1) \quad \forall (u, v) \in E$$

- The above constraint ensures that at least one of the vertices u or v is in the vertex cover.
- But those constraints are not linear. Instead we can use the following linear constraints:

$$x_u + x_v \geq 1 \quad \forall (u, v) \in E$$

- The remaining (non-linear) constraints are:

$$x_v \in \{0, 1\} \quad \forall v \in V$$

- Up to this point our formulation (integer linear programming) is equivalent to the original problem.

Linear programming approximation

- We can **relax** the constraints $x_v \in \{0, 1\}$ to $0 \leq \bar{x}_v \leq 1$.
- The resulting system is a **linear programming approximation** of the original problem

$$\begin{array}{ll}\text{minimise } \bar{z} = \sum_{v \in V} \bar{x}_v \cdot w(v) & \\ \text{subject to } \bar{x}_u + \bar{x}_v \geq 1 & \forall (u, v) \in E \\ \bar{x}_v \leq 1 & \forall v \in V \\ \bar{x}_v \geq 0 & \forall v \in V\end{array}$$

- Where we have used $\bar{x}_v$ distinguish the relaxation from the original problem.
- We know how to solve such problems efficiently.

- The solution to the linear programming relaxation may not be integral, i.e. $\bar{x}_v$ may not be 0 or 1.
- In such a case how do we decide if a vertex is in the vertex cover?
- We use the following rounding rule:

$$x_v = \begin{cases} 1 & \text{if } \bar{x}_v \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

- Note that this rule ensures that we obtain a valid solution (not necessarily optimal).
- This is because the constraint $\bar{x}_u + \bar{x}_v \geq 1 \quad \forall (u, v) \in E$ implies that $\bar{x}_u$ and $\bar{x}_v$ cannot be both < 0.5 and therefore either $x_u = 1$ or $x_v = 1$ (or both) in C for every edge $(u, v) \in E$.

Finding a lower bound

- Let $\bar{z}^*$ and z^* be the optimal value of the linear programming relaxation and the original problem respectively.
- Recall that the linear programming relaxation is obtained by removing integral constraints from original problem
- This means that the **feasible region** of the linear programming **relaxation is a superset** of the feasible region of the original problem.
- Therefore $\bar{z}^* \leq z^*$ and the solution to the linear programming relaxation is a lower bound on the optimal value of the original problem.
- Let x_v^* be the solution obtained by rounding the LP relaxation solution.
- And $\bar{x}_v^*$ be the solution to the LP relaxation.

LP relaxation solution is a 2-approximation

- We show that the solution to the linear programming relaxation is a 2-approximation of the original problem.

$$\sum_{v \in C} w(v) \leq \sum_{v \in C} 2 \cdot \bar{x}_v^* \cdot w(v) \quad (1)$$

$$\leq \sum_{v \in V} 2 \cdot \bar{x}_v^* \cdot w(v) \quad (2)$$

$$= 2 \cdot \bar{z}^* \quad (3)$$

$$\leq 2 \cdot z^* \quad (4)$$

- (1) is true because $C = \{v \in V : \bar{x}_v \geq 0.5\}$
- (2) is true because $\bar{x}_v \geq 0$ for all $v \in V$.
- (3) is true because $\bar{z}^*$ is optimal solution for the LP relaxation.
- Finally (4) is true because $\bar{z}^*$ is a lower bound for z^*